

Pelagic Species Trait Database

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Description

Biological traits are increasingly used to characterize predator-prey interactions within changing ecosystems (McGill et al. 2006). When combined, a suite of traits are useful to describe diet selection (Green & Côté 2014) or prey guilds based on functional role (Hardy et al. accepted) to better predict interactions within ecological communities, especially in the scope of global change. In particular, habitat, behavior, morphology, and nutritional quality are important traits that can affect prey vulnerability across different aspects of the predation process (Pyke et al. 1977) (encounter, attack, capture). Habitat use (e.g., water column position) and migration behaviors impact encounter rates through spatiotemporal overlap, and schooling behavior can deter or facilitate predator attack. Morphological traits such as body shape and physical defenses influence the costs of prey capture, while body size affects consumption for gape-limited predators, and relative eye, fin or appendage size can influence predator detection and evasion (Green et al. 2019). Nutritional quality traits also mediate prey selection since predators select prey items in a manner that maximizes energy gain while minimizing energy expenditure (Pyke et al. 1977). Nutritional quality varies not only among species but also within species, reflecting

geographic, seasonal, interannual, and longer-scale changes in environmental conditions (Shalders et al. 2022).

Understanding how species will interact with one another is important for predicting how ecological systems and services will be altered by forces such as climate change and biological invasions (Poloczanska et al. 2016, Bartley et al. 2019). Trait-based approaches focus on the mechanistic drivers of ecological interactions and are emerging as a useful method for predicting variability in species distributions, community structures, and population dynamics under global change (Laigle et al. 2018, Brose et al. 2019, Green et al. 2022). Further, identifying traits that recur across unrelated prey taxa offer a means to better anticipate predator resource use by simplifying complex foraging dynamics (Hardy et al. accepted). Assembling comprehensive databases of traits for biological communities facilitates ecological modeling of future species abundances, distributions, and food web structures (Gallagher et al. 2020, Green et al. 2022).

This dataset contains traits for adults, juveniles, and larvae of 529 pelagic fish and invertebrate species found worldwide. Traits included describe 1) habitat use and behavior, 2) morphology and morphometrics, 3) nutritional quality (lipid, protein, energy density), and 4) population status information. The dataset was specifically created for its application in multi-faceted ecological modeling occurring in the California Current system (CCS) located within the NE Pacific Ocean. Therefore, species in the dataset are primarily from the CCS and broader NE Pacific Ocean to encompass both known prey and potential prey for pelagic predators (Hardy et al. accepted) (given anticipated future shifts in species distributions). Globally important pelagic species known to be consumed by top ocean predators that are found in both the NE Pacific and other ocean basins (NW Pacific, Atlantic, Indian, Mediterranean) are also included to promote the use of trait-based approaches in marine ecosystems and predator populations worldwide. Detailed protocols are provided for trait data collection to serve as a framework for the expansion of this dataset in the future for other systems and predators.

The database is released under a CC-BY-NC license permitting reuse for non-commercial purposes with citation of this database, the data descriptor publication (Gleiber et al. in review) and any original sources, when possible. Users are requested to provide contact information prior to downloading to ensure updated versions are distributed to the user community, as well as enable solicitation of feedback from the community on the database design for user accessibility. New versions of the database will be released as data gaps are filled and values are updated using the methods described above.

Tables (*See the separate metadata file associated with each table for more information*)

Table 1: Habitat, behavioral, morphological, nutritional, and population status trait variables for adults (n = 498), juveniles (n = 469), and larvae (n = 409) of pelagic species. NA = species/lifestage searched with no information found for the trait variable, -9999 = trait variable not searched for the species/lifestage. See Tables 3-8 for data sources, and Tables 9-11 for data collection information.

Table 2: Source of species lists used to select species included in the database.

Habitat/behavioral traits:

Table 3: Habitat and behavioral traits for pelagic species. Includes source information for trait data as well as ordinal and binary versions of some variables.

Morphological traits:

Table 4: Morphological traits & morphometric ratios for pelagic species. Includes source information for morphological trait data, as well as ordinal and binary versions of some variables. NA = species/lifestage searched with no information found for the trait variable, -9999 = trait variable not searched for the species/lifestage. Morphometric ratios are mean values calculated from replicate images per species and life stage; sample sizes and standard deviations are reported. Morphometric ratios were not collected for larvae. Source information for individual morphometric ratios are in Table 5.

Table 5: Individual relative body measurements and morphometric ratios for species with image data collection information and sources.

Nutritional quality traits:

Table 6: Mean nutritional quality traits (lipid, protein, and energy density) for each species. Mean values are calculated separately for 1) all global regions, and 2) only values the Pacific Ocean. Sample sizes and standard deviations are reported. Nutritional quality data is integrated across adult and juvenile lifestage due to low sample size and absence of lifestage information in the literature.

Table 7: Nutritional quality information for lipid, protein, and energy density content and associated metadata for pelagic species from published literature and unpublished data.

Population status traits:

Table 8: Population status traits for pelagic species. Includes source information for trait data, as well as ordinal and binary versions of some variables.

Data collection information:

Table 9: Initially collected trait data of habitat, behavioral, morphological, and population status variables. Provided as an example framework for future data collections. Metadata includes detailed instructions on data collection for each trait variable. Data is processed using the R code (*'Pelagic_Species_Trait_Database.Rmd'*) included in the database to create Tables 1,3,4,8.

Table 10: Initially collected morphometric measurement data from species images. Provided as an example framework for future data collections. Metadata, the *'Morphometric Ratios Data Collection Protocol'*, and *'Morphometric Measurement Guide'* includes detailed instructions on data collection for each variable. Data is processed using the R code (*'Pelagic_Species_Trait_Database.Rmd'*) included in the database to create Tables 1,4,5.

Table 11: Initially collected nutritional quality data from published literature and unpublished data. Metadata and the *'Nutritional Quality Data Collection Protocol'* includes detailed instructions on data collection for each variable. Data is processed using the R code (*'Pelagic_Species_Trait_Database.Rmd'*) included in the database to create Tables 1,6,7.

Methods

Species list

The Pelagic Species Trait Database (Table 1) includes species representing pelagic communities of the CCS, as well as cosmopolitan species known to be important prey for pelagic predators in other ocean basins (n = 529; Table 2). For the NE Pacific, we included species observed in 15 years (2005-2019) of annual NOAA midwater trawls conducted in the CCS conducted by the Southwest Fisheries Science Center Fisheries Ecology Division (SWFSC-FED) Rockfish Recruitment and Ecosystem Assessment Survey (Sakuma et al. 2016, Santora et al. 2021), SWFSC Fisheries Resources Division (SWFSC-FRD) California Current Ecosystem Survey (Zwolinski et al. 2019), Northwest Fisheries Science Center Fish Ecology Division (NWFSC-FED) Stock Assessment Improvement Program (2005-2011) (Phillips et al. 2009), and NWFSC Coastwide Cooperative Pre-Recruit Survey (2011-2019) (Brodeur et al. 2019). To encompass other communities in the NE Pacific we included species sampled by Fisheries and Oceans Canada (DFO; 2017 - 2019) and the North Pacific Anadromous Fish Commission's International Year of the Salmon (2020). Beyond the NE Pacific, we included many known prey of a highly-migratory generalist predator, albacore tuna (*Thunnus alalunga*), compiled from a recent global meta-analysis of its diet (Hardy et al. accepted) (1880 - 2020). We note the dataset includes all species reported in the diets of *T. alalunga* collected in the CCS from 2005-2019 (Glaser 2010, Glaser et al. 2015, Madigan et al. 2015, Nickels et al. 2023). Overall, species in this dataset represent 118 families of fish, 27 families of cephalopods, and 66 families of other invertebrates (e.g., crustaceans, jellies). Species names and phylogenetic information were verified using the Open Tree of Life (OpenTreeofLife et al. 2019) and the World Register of Marine Species (www.marinespecies.org).

Trait Data Collection

For each species we collected information on four trait categories: (1) habitat/behavior, (2) morphology (including morphometric ratios), (3) nutritional quality, and (4) population status (Tables 1, 3-8). We searched online repositories (FishBase [www.fishbase.org] and SeaLifeBase [www.sealifebase.ca], Google Images [www.images.google.com]) and primary literature through bibliographic databases (Google Scholar [www.scholar.google.com], Web of Science [www.webofscience.com], Aquatic Sciences and Fisheries Abstracts [https://proquest.libguides.com/asfa], Federal Science Library Canada [https://science-libraries.canada.ca] for species-level information and images. Search terms included scientific name or common name, lifestage (e.g. adult, juvenile, larva), and the trait of interest. When possible, trait data collection protocols followed standards outlined in the Preferred Reporting Items for Systematic reviews and Meta-Analyses (O’Dea et al. 2021), including consistent search terms, eligibility criteria for including data sources, data collection, source metadata, review, and bias reporting. We note that known trait information may change after data collection, especially for cryptic species and/or lifestages. All data manipulations, calculations, and summaries are described below and detailed in the R code included with the dataset (*‘Pelagic_Species_Trait_Database.Rmd’*). Table 1 summarizes all key trait variables for each species and lifestage. Trait source information, notes, expanded variables, and individual replicates for mean values are detailed by trait type in Tables 3-8.

Habitat/behavioral traits:

Habitat/behavioral traits include vertical and horizontal habitat (categorical), depth and temperature range (numeric), aggregation, diel vertical migration, seasonal migration, and refuge use behaviors (Table 1, 3). These traits were collected separately for adult, juvenile, and larval lifestages, and qualitative trait confidence level was noted based on the amount of available sources (Table 3). If limited information was available for the species, adult traits were applied to the juveniles (but not larvae), unless specific information was found indicating a different value and it was reasonable based on both lifestages likely occupying the same habitats (Hardy et al. accepted). Vertical and horizontal habitat use traits were directly recorded from online repositories and corroborated with species distribution maps and reported depth ranges from the primary literature when possible (Hardy et al. accepted). Where published literature expanded on, or differed from a general value reported by repositories, we used values from the published literature and data. For some trait variables, ordinal and binary versions were also included to facilitate future analyses.

Morphological traits and morphometric ratios:

Morphological traits include lifestage-specific length range, body shape, and the presence and nature of defensive spines, exoskeleton, transparency, disruptive coloration patterns, silvering, countershading, and photophores (Table 1, 4). Morphometric ratios (relationships between body dimensions) are also included with the morphological traits to describe different aspects of body shape using continuous, numerical data, and were part of a separate data collection effort, which is detailed below (Table 1, 4, 5). Morphometric ratios were only collected for adults and juveniles. Ratios were also used to convert different length types to total length, thus larvae lengths were unable to be converted to total length in some cases (*'Morphometric Measurement Guide'*, *'Pelagic_Species_Trait_Database.Rmd'*).

For each species and lifestage we quantified the relative total length (TL), standard length (SL), total height (TH), body height (BH), eye diameter, and dorsal fin height (see *'Morphometric Measurement Guide'*). Measurements were taken from ~6 replicate images (range: 1-10) that were selected from the image search results based on a set of criteria to ensure accurate relative measurements. The criteria includes that the images show: i) the correct species and lifestage, ii) the organism perpendicular to the frame of reference and not angled toward/away from the camera, iii) all dimensions measurable from the same image (lateral view for most organisms, dorsal view for flatfish, rays, crabs), and iv) soft-bodied organisms (e.g., cephalopods) with arms extended. When photographs or drawings from literature sources were not found we used the best available images. For some rare species and juvenile lifestages, the selection of images to choose from was limited and we were not able to adhere to all the criteria. Measurements based on any images that do not meet all the measurement criteria are noted (see *'Morphometric Trait Data Collection Protocol'*). While morphometric ratios were lifestage-specific, we acknowledge that there is likely some variation within a lifestage that we are not capturing based on sample sizes and images available.

Relative measurements for each image were collected in pixels, using ImageJ (Schneider et al. 2012) and measurements for each dimension were based on definitions from the literature (*'Morphometric Measurement Guide'*). Since this type of dataset was initially developed for fish (Green & Côté 2014), SL was converted from other length types (e.g., mantle length, shell length, carapace width) depending on the taxa (*'Morphometric Measurement Guide'*, *'Pelagic_Species_Trait_Database.Rmd'*). Morphometric ratios were calculated for each image as TL:SL, TL:TH, SL:TH, TL:BH, SL:BH, and eye diameter:TL (Table 5). Fins can be folded or destroyed when individuals are removed from the water and/or preserved, and so fins were not always visible. In these instances, TH (includes dorsal and anal fins) or TL (includes caudal fin) could not be measured. Similarly, TL could not be measured if the arms (for cephalopod) or urosome (for crustaceans) were folded or not visible. Mean ratios were then calculated for each species and lifestage ($n = 2-10$; Table 1, 4), except in instances where only a single image was available. If species-specific morphometric ratios were not available, a proxy for the next available level of taxonomic identification (e.g genus, family) was used for the lifestage. Trait source information and notes on data collection are reported (Table 5).

Nutritional quality:

For each species, we quantified lipid content (%), protein content (%), and energy density (kJ/g) through a meta-analysis of published literature (Table 1, 6, 7). Keyword search terms include the

scientific name, each nutritional quality metric (or synonym), and optional location keywords (see below). To expand the search, we excluded quotations on some search terms to allow the search engine to also return results with synonyms (e.g., *percent* includes results for *proportion*). We also include data from laboratory analyses of energy density by bomb calorimetry and lipid and protein percentages by proximate composition for specimens collected in the CCS (see Supplementary Materials). Search results were evaluated for relevance using the title, abstract, keyword searches within the publication, and/or by visually scanning the paper.

Quality metric	Keywords	Location keywords (optional)
Lipid Content	“Genus species” “lipid” percent “Genus species” lipid percent “Genus species” “fat” percent “Genus species” fat percent	“California”, “Oregon”, “Washington”, “Baja”, “British Columbia”, “Alaska”, “Bering Sea”, “Japan”, “Kuroshio”, “Pacific”
Protein Content	“Genus species” “protein” percent	
Energy Density	“Genus species” “energy” density “Genus species” “energy” content “Genus species” “calorie” content	
Lipid & Protein	“Genus species” “proximate” composition	

Lipid, protein, and energy density information were recorded to the highest level of detail reported in the publication, using individual values instead of mean values when possible (Table 7). We recorded lipid and protein content in percent weight, and energy density in kJ/g, converting units as necessary. Research articles predominantly reported nutritional quality content as a proportion of wet weight (ww), however dry weight (dw) and ash-free dry weight (afdww) data are also included in the literature. We standardized nutritional quality metrics as wet weight, converting dry weight and ash-free dry weight as follows (*‘Pelagic_Species_Trait_Database.Rmd’*):

$$\begin{aligned} \% \text{ ww} &= (\% \text{ dw} / 100) * (100 - \% \text{ water}) \\ \% \text{ ww} &= (((\% \text{ afdw} / 100) * (100 - \% \text{ ash dry weight})) / 100) * (100 - \% \text{ water}) \end{aligned}$$

Conversions use water (or moisture) content (%) and/or ash content (% dry weight) associated with the nutritional quality data reported in the paper. If these values were not reported with the dw or afdw values, the nutritional quality data was reported (Table 7), but percentage wet weight could not be calculated. However, a water content proxy from the same species in the same region was used in a few invertebrates with consistently high water content (e.g., pyrosomes) or if a proxy has been used in the literature (e.g., krill).

In addition to nutritional quality information, we also reported covariates associated with each data point, when available. Covariates included sample size, moisture content, ash content, lifestage, age, sex, wet weight (mean, minimum, maximum), length (mean, minimum, maximum), temporal variables (sampling start/end year, month, day), and geographic location (latitude, longitude, ocean basin, descriptive location). Mean wet weight and length were estimated from the minimum and maximum values if not directly reported. Additionally, some location covariates were estimated if not reported (e.g., coordinates estimated from descriptive location), and details were noted. We standardized location information using Longhurst provinces, assigned by intersecting coordinates with Longhurst province polygons in R software (R Core Team 2022) (Version 3.6.0), or manually if only a general location was reported.

Mean nutritional quality metrics were calculated for each species using data (standardized to percentage wet weight) from 1) all global regions and 2) only values from the Pacific Ocean (Table 1, 6). We also include disaggregated, individual nutritional quality data in the dataset (Table 7). Due to limited information about lifestage or age included in the literature, nutritional quality values from adults and juveniles are combined in mean values (Table 1, 6). Nutritional quality data was not collected for larvae, thus mean nutritional quality values were only applied to adults and juveniles (Table 1).

Population status:

Population status traits include trophic level, fisheries and conservation status (Table 1, 8). The trophic level is an estimate collected from FishBase and SealifeBase calculated by Ecopath software using the trophic levels of a predator's known prey (Pauly & Christensen 1995, Pauly et al. 1998). Trophic level is only available for adult lifestages of all fish species and some marine invertebrates. The fishery status was primarily collected from FishBase/SealifeBase, as well as online keyword searches for the species scientific or common name and the term “commercial”, “recreational” or “fishery”. The conservation status is collected from the IUCN Red List of Threatened Species (www.iucnredlist.org) for adults only.

Data Collection Instructions

For all trait data we provide data collection instructions and materials to serve as a framework for the expansion of this database in the future for other systems and predators. Files detailing trait data collection are listed below in the order they should be opened for instruction.

‘Pelagic_Species_Trait_Database.Rmd’ - Instructional R code to provide a framework for the creation of future trait databases. We aim for users to be able to replicate the data collection protocols, processing, and summaries included in the Pelagic Forage Species Trait Database. All input files and generated output files can be downloaded from the database on Borealis.

Habitat, behavior, morphology, & population status traits:

Table 9 metadata (*data_collection_9metadata_traits*) - detailed instructions for collection of each variable including categorical values, suggested sources, keyword search terms, additional materials, and functions/fields from the ‘rfishbase’ package for automated data extraction from FishBase/SeaLifeBase (*‘Pelagic_Species_Trait_Database.Rmd’*).

Table 9 (*data_collection_9_traits*) - trait data collection spreadsheet.

Morphometric Ratios:

‘Morphometric Ratios Trait Data Collection Protocol’ - step-by-step instructions on building a data library of body shape ratios (body length : body height, standard length : total length, etc.).

‘Morphometric Measurement Guide’ - illustrated instructions on collecting body measurements for fish, cephalopods, crustaceans, and other marine invertebrates. Includes definitions of measurements used in the database.

Table 10 (*data_collection_10_morphometric_ratios_data*) - trait data collection spreadsheet.

Table 10 metadata (*data_collection_10metadata_morphometric_ratios*) - instructions for collection of each variable (and metainformation).

Nutritional Quality:

‘Nutritional Quality Data Collection Protocol’ - step-by-step instructions on building a data library of nutritional quality information (lipid, protein, energy density content).

Table 11 (*data_collection_11_nutritional_quality*) - trait data collection spreadsheet.

Table 11 metadata (*data_collection_11metadata_nutritional_quality*) - instructions for collection of each variable (and metainformation).

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